

CRYSS Input Parser

Methodology and Tolerance Handling

CRYSS Project

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1 Purpose and Scope

The input parser turns a workbook of experimental records into a set of *mixtures*, each holding several *slices* (i.e. composition slices at specified recoveries), ready for the optimizer. It stops at the parser/optimizer boundary. This document describes what the code in `input_parser/` actually does, in particular how the three tolerances are used and why. Mixtures are never merged or averaged against each other, and slices of one mixture are never averaged into a single R .

Guiding principle. Every retained row is a measurement that constrains the model, so the parser rejects only rows that are demonstrably wrong. A row with scattered composition whose R is still consistent is kept. Rejection is intended for wild outliers, and it becomes possible only when there are enough replicate rows to identify them.

2 Data Model

Each record (row) carries a mixture identifier, the starting composition X_c , the solid composition X_s , the liquor composition X_l , and a relative solvent volume. All compositions are mole or mass fraction vectors over the n components.

- A **mixture** is the set of rows sharing a `mixture_id` (and the same salt/derivative label, see below).
- A **slice** is one solvent-volume point within a mixture. Each slice has its own R . Different slices have different R values by design and all are passed to the optimizer.
- A **replicate** is a repeat measurement of the same slice. Replicates are the only rows that are compared with each other.

3 The Three Tolerances

Name	GUI control	Default	Role
τ_{X_c} (XC_ANALYTICAL_TOL)	Analytical error	2%	Starting compositions X_c of rows in one mixture must agree within this value.
τ_{noise} (BASE_TOL)	Noise	5%	Scatter allowed in X_s and X_l between replicate rows (cross contamination). Expected to exceed τ_{X_c} .
τ_R (dynamic)	none (derived)	7.3% at defaults	Allowed spread in R between replicates. Computed from the other two.

The GUI writes the two spin-box values into the parser modules (`mixture_set_detector` and `slice_builder`) before each parse, and restores the defaults afterwards.

3.1 Derived R tolerance

τ_R is not a fixed constant. It is recomputed whenever it is used from the current τ_{X_c} and τ_{noise} :

$$\tau_R = \sqrt{\tau_{X_c}^2 + 2\tau_{\text{noise}}^2}. \quad (1)$$

The lever-rule value of R depends on X_c , X_s and X_l . Equation (1) is a quadrature sum of one analytical contribution and two noise contributions, taken as independent. It is a pragmatic bound rather than a rigorous error propagation through the lever rule, which also depends on how close X_s and X_l are. At the defaults ($\tau_{X_c} = 2\%$, $\tau_{\text{noise}} = 5\%$) $\tau_R = 7.3\%$. The former fixed value was 8%.

4 Processing Pipeline

4.1 Loading and pre-filters

1. Records are read from the NODES format or from the base-case CRYSS workbook. Rows without explicit X_c cannot be used and are dropped, because X_c is not be inferred from X_s and X_l (in the case where user measures only X_s and X_l , and by w/w determination determines the mass of material dissolved in the solution phase, i.e. liquor analysis with no solid separation/analysis, the recovery may be calculated and X_s determined via the lever rule, it is expected that the X_s are calculated and provided in the input file).
2. **Salt consistency.** Within a mixture, if some rows carry a salt/derivative label and others are blank, the blank rows are rejected. Mixtures with several distinct labels are split into separate groups.
3. **Unconstrained rows.** A row whose solid collapses to a single component ($\max X_s \geq 0.99$ with all other components $\leq 10^{-6}$) is rejected as “Rmax – 0% unconstrained” (this region is outside the eutectic and has no predictive value in the optimizer).

4.2 Row R from the lever rule

For each component i of a row,

$$R_i = \frac{X_{c,i} - X_{l,i}}{X_{s,i} - X_{l,i}}. \quad (2)$$

Components with $|X_{s,i} - X_{l,i}| < 10^{-12}$ are skipped, since the quotient is undefined. Outliers among the component values are then rejected within that row: components farther than 2 MAD from the row median are dropped (no rejection if the MAD is zero), and the row R is the mean of the remaining values. This is the per-row outlier rejection. A component with nearly equal X_s and X_l can give a very large R_i (values near 26 occur) and is removed here.

4.3 Slice assignment

In NODES format, rows within a mixture are assigned a slice number from their relative solvent volume: rows with the same volume (within 10^{-6}) form one slice. For CRYSS format, rows no solvent volume is supplied and are placed by X_s/X_l similarity with tolerance $\min(\tau_{\text{noise}}, 4\%)$. Slice assignment never rejects a row.

4.4 Replicate groups and the row filters

Rows in the same slice are treated as replicates. The filters below are applied in order. Each rejected row is recorded in a rejection ledger with its mixture, slice and reason.

1. **R outlier pre-filter.** Let $\tilde{R}$ be the median row R of the slice and MAD the median absolute deviation. A row is rejected if

$$|R - \tilde{R}| > \max(2 \text{MAD}, \tau_R). \quad (3)$$

Taking the larger of the two prevents a tight replicate group from shrinking the band below the measurement noise and discarding good rows. Only rows well outside the tolerance are dropped. Reason: *R outlier within slice*.

2. **Xc agreement (τ_{X_c}).** The canonical mixture composition $\bar{X}_c$ is the mean of the rows after removing rows more than 2 MAD from the median in any component. A row is rejected if any component satisfies $|X_{c,i} - \bar{X}_{c,i}| > \tau_{X_c}$. This guards against re-analysis, instrument error and sample inhomogeneity in the starting composition. Reason: *Xc outside analytical tolerance*.
3. **Xs/Xl scatter (τ_{noise}), gated by R.** For each slice with replicates, the per-component median X_s and X_l are formed. A row is rejected only if *both* hold:
 - some component of X_s or X_l differs from the slice median by more than τ_{noise} ; and
 - its R differs from the slice median R by more than τ_R .

Scatter in X_s or X_l alone does not reject a row if the resulting R agrees. Reason: *Xs/Xl outside noise tolerance*.

4. **Final replicate check.** Rows whose R cannot be computed are rejected, and so are rows that exceed Equation (3) or fail the X_c check against the canonical composition.

Rows that survive are passed on. A mixture with a single row bypasses the replicate checks.

4.5 Slice records handed to the optimizer

For each slice the parser stores the canonical values

$$\tilde{X}_s = \text{median}_k X_s^{(k)}, \quad \tilde{X}_l = \text{median}_k X_l^{(k)}, \quad \tilde{R} = \text{median}_k R^{(k)}, \quad (4)$$

together with the canonical X_c of the mixture and the retained raw rows. The medians are over replicates of the *same slice* only. There is no mixture-level R (only slice-level R are meaningful).

5 Reporting

All rejections are recorded in one ledger, and the summary reads its counts from it. The summary therefore always reports a record count and reason counts that agree. The reasons are:

- Rmax – 0% unconstrained
- below valid-band threshold (low R relative to the slice)
- R outlier within slice
- X_c outside analytical tolerance
- X_s/X_l outside noise tolerance
- R could not be computed

The QC trace header lists τ_{X_c} , τ_{noise} and the derived τ_R in percent with one decimal place.

6 Observed Behaviour

The table gives the total number of rows rejected on four test workbooks at several tolerance settings. The three error-baked workbooks carry Gaussian error in X_s and X_l .

Workbook (rows)	2/5	2/3	2/2	2/1	1/1
Clean Vol03 (165)	0	0	0	0	0
Vol03, 5% error (165)	0	0	0	0	1
Vol05, 5% error (29)	0	0	0	0	2
Vol05, 7% error, replicates (72)	0	0	0	3	10

Column headings give analytical/noise in percent. Key observations:

- At the defaults and down to 2% noise nothing is rejected, so all rows reach the optimizer.
- Before the $\max(2 \text{ MAD}, \tau_R)$ change, a 2 MAD rule on its own removed one replicate in 15 of 25 slices irrespective of tolerances, sometimes with an R deviation of only 0.4%.
- Before the R gate on the X_s/X_l check, tightening noise to 1% removed 28 rows from the 7% workbook, several with consistent R . With the gate the count is 3.

7 Known Limitations

- **Single-row slices are never tested.** Every comparison is against the slice median, so a lone row always agrees with itself. The X_s/X_l and R -outlier filters have an effect only where replicates exist.
- **Two-row slices are symmetric.** The median is the midpoint, so both rows deviate equally, and both pass or both fail. Identifying the bad row requires three or more replicates.
- **Equation (1) is heuristic.** It does not account for the $X_s - X_l$ denominator in Equation (2).
- **Defaults are untested against deliberately wild rows.** The test workbooks contain noisy but consistent data. A workbook with a few rows pushed 15–20% off in R would show whether the default noise value lets a wild row through.
- **Replicate numbering.** Rows in a slice are numbered into replicate groups using τ_{noise} on X_s and X_l . This labels rows and does not reject them.